PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	AN INTERDISCIPLINARY SYSTEMATIC REVIEW: DOES ALIGNMENT BETWEEN SYSTEM AND DESIGN SHAPE ADOPTION AND USE OF BARCODE MEDICATION ADMINISTRATION TECHNOLOGY?
AUTHORS	Williams, Rachel; Aldakhil, Reham; Blandford, Ann; Jani, Yogini

VERSION 1 – REVIEW

REVIEWER	Blagoj Ristevski Faculty of Information and Communication Technologies - Bitola, "St. Kliment Ohridski" University - Bitola, Republic of Macedonia
REVIEW RETURNED	02-Oct-2020

GENERAL COMMENTS	Authors systematically searched for literature produced between April 2000-April 2020, particularly the Cumulative Index of Nursing, and Allied Health literature, PubMed, OVID MEDLINE and Google scholar in order to describe how human factors related determinants for BCMA have been researched and reported by healthcare and HCI disciplines. Despite the fact that this manuscript should represent an interdisciplinary systematic review, the methodology is vague, the results are not presented in appropriate tables (with numbers) as well there are no suitable charts of this review. Concluding remarks and directions for further works should be explained in a more precisely manner.

REVIEWER	Dr. Vanessa Burkoski, RN, BScN, MScN, DHA Humber River Hospital Toronto, Ontario Canada
REVIEW RETURNED	28-Oct-2020

GENERAL COMMENTS	5) Although no patient involvement was declared, a statement regarding 'no personal or identifying private health information would be derived from the public sources being searched" may provide further clarity. Timely and important topic.

REVIEWER	Ross Koppel Univ of Pennsylvania
REVIEW RETURNED	20-Dec-2020

GENERAL COMMENTS	I have an odd and difficult statement. While the paper is generally excellent, they have made serious misstatements about my own research (Koppel et al, 2008), suggesting they didn't read beyond

	the abstract. They say the work is based on about 60 observations, but that's incorrect. As stated in the article, it was based on careful analysis of 307,698 scans of medication administrations. Obviously, that's more than 60. Also, we did not say most workarounds present dangers. Certainly some do, but almost all workarounds are made clinicians seeking to help patients and avoid barriers created by poor design of the technology and workflow. As to their summaries of others' work, they seem fine. Again, I emphasize that the paper is generally excellent and a valuable contribution. I am not happy making a judgement from an N of 1. But I happen to know that work rather well, and am thus aware of the distortions of it in the paper. Note that assuming this is their only error, the paper is easily corrected.

REVIEWER	Susan Moffatt-Bruce Royal College of Physician and Surgeons of Canada
REVIEW RETURNED	21-Dec-2020

GENERAL COMMENTS	The paper is very well written and has tremendous depth. I do think there is an opportunity to actually state what the next steps are and how a human factors approach would be potentially introduced and tested.

VERSION 1 – AUTHOR RESPONSE

Response for reviewer 1.
Prof. Blagoj Ristevski, Saint Kliment Ohridski University Bitola, Faculty of Information and Communication Technologies Bitola
Thank you for taking the time to review our paper and for your feedback. The PRISMA checklist approach was used to guide methodology, the tables are numbered appropriately and the PRISMA flow chart is included as figure 1 to demonstrate the selection process of the papers.
Directions for further work has been expanded to include more detail and specificity.

Response for reviewer 2
Dr. Vanessa Burkoski, Humber River Hospital
Thank you for your feedback, the PPI statement has been updated in line with your recommendation to increase clarity.

Response for reviewer 3
Dr. Ross Koppel, Univ of Pennsylvania
Thank you for your feedback and apologies for the misunderstanding. The 62 observations in the table referred specifically to the qualitative element of this study. However, we also referred to the quantitative elements in the review itself. We have now updated the table and text to reflect this.
The paper has been adjusted to reflect your stated outcome, that not all workarounds are dangerous.

Response for reviewer 4.
Dr. Susan Moffatt-Bruce, The Ohio State University College of Medicine
Thank you for your feedback, the paper has been amended to include details of future work.

VERSION 2 – REVIEW

REVIEWER	Blagoj Ristevski Faculty of Information and Communication Technologies - Bitola, "St. Kliment Ohridski" University - Bitola, Republic of Macedonia
REVIEW RETURNED	28-Feb-2021

GENERAL COMMENTS	The list of the references should be updated - only a few papers are published in the past 3 years.

REVIEWER	Ross Koppel Univ of Pennsylvania
REVIEW RETURNED	18-Feb-2021

GENERAL COMMENTS	This paper has much to recommend it. There are still, however, some blind spots that must be addressed. Example: the abstract sentence reads: Barcode medication administration (BCMA) is increasingly utilised to improve safety in healthcare. However, how human factors influence adoption and usability of this technology is relatively unknown. Problem: 1. As the authors find, BCMA is used IN THE HOPE OF IMPROVING SAFETY. The authors note the data are still out. They should change that sentence. Also, They repeat the line several times that there is little research or known findings about human-computer interactions (HCI) on complex institutions This is nonsense. It ignores both their own findings and over 50 years of human factors (HF) research. This refusal to notice the many decades of research on HF and HCI Problem 2. Their sentence in the discussion is just wrong. The wrote: "In particular, very little prior research has explored the interdependencies between technology design and use within complex systems (a rare exception being the work of (38)). More specifically, little work in Human–Computer Interaction, which addresses design for usability, utility and safety, has focused on how to design technology to support complex tasks." That's just absurd. Again, their own research and 50 years of literature contradicts this finding. They have done a fine job of presenting the literature on BCMA. It's therefore foolish to insist that they are somehow unique in this effort. .

VERSION 2 – AUTHOR RESPONSE

Reviewer: 3

Dr. Ross Koppel, Univ of Pennsylvania

Comments to the Author:

This paper has much to recommend it. There are still, however, some blind spots that must be addressed.

Example: the abstract sentence reads: Barcode medication administration (BCMA) is increasingly utilised to improve safety in healthcare. However, how human factors influence adoption and usability of this technology is relatively unknown.

Problem: 1. As the authors find, BCMA is used IN THE HOPE OF IMPROVING SAFETY. The authors note the data are still out. They should change that sentence.

Also, They repeat the line several times that there is little research or known findings about human-computer interactions (HCI) on complex institutions This is nonsense. It ignores both their own findings and over 50 years of human factors (HF) research.

This refusal to notice the many decades of research on HF and HCI Problem 2. Their sentence in the discussion is just wrong. The wrote: "In particular, very little prior research has explored the interdependencies between technology design and use within complex systems (a rare exception being the work of (38)). More specifically, little work in Human–Computer Interaction, which addresses design for usability, utility and safety, has focused on how to design technology to support complex tasks." That's just absurd. Again, their own research and 50 years of literature contradicts this finding. They have done a fine job of presenting the literature on BCMA. It's therefore foolish to insist that they are somehow unique in this effort. .

Response to reviewer 3.

Thank you for your feedback.

We have updated the abstract to more clearly reflect the contents of the review.

We are of course aware of the immense amount of work that has already been done in this field; we were trying to convey something much more subtle about the challenge of explicitly designing technology for complexity, but clearly failed in that. Since this was never intended as the core focus of the paper, we have removed the offending sentences.

Reviewer: 1

Prof. Blagoj Ristevski, Saint Kliment Ohridski University Bitola, Faculty of Information and Communication Technologies Bitola

Comments to the Author:

The list of the references should be updated - only a few papers are published in the past 3 years.

Response to reviewer 1

Thank you for your feedback.

The systematic review covers the period April 2000-April 2020, as reported. It would be methodologically unsound to retrospectively add more recent papers to this. We have conducted a

further search for relevant literature (May 2021) and added discussion of three further relevant studies.